

Docker Network Connectivity Issue SOP

Objective

Restore container-to-container or container-to-host connectivity when Docker networking, bridge configuration, DNS, or port mapping fails.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T008
Issue	Docker network issue
Category	Networking
Service	Docker
Severity	Medium
Status	Resolved
Description	Containers cannot communicate with each other
Expected Resolution	Verify docker network settings and bridge configuration

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L018	5/1/2026 10:09	WARNING	Network connectivity issue	DOCKER_NET
L038	5/1/2026 10:19	WARNING	Network connectivity issue	DOCKER_NET

Procedure

- Confirm source container, destination container, target host, port, protocol, and exact connectivity error.
- Check container status and ensure both source and destination containers are running.
- Inspect Docker networks and verify both containers are attached to the expected network.
- Validate published ports, expose settings, container DNS names, and service aliases.
- Test connectivity using ping, curl, nc, or wget from inside the source container.
- Review firewall, host routing, bridge network, and compose network configuration.
- Attach containers to the correct user-defined network or fix port mapping.
- Retest service communication and monitor error logs for recurrence.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Containers cannot communicate	Different Docker networks	Attach containers to same network
Connection refused	Service not listening on target port	Fix app bind address or port
DNS name not resolving	Default bridge limitation or alias issue	Use user-defined network and service name
Host cannot reach container	Port not published	Map host port to container port

Best Practices

1. Use user-defined bridge networks for multi-container apps.
2. Document container ports and service names.
3. Avoid relying on container IPs in configs.
4. Add network checks to deployment validation.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook